



RIVER VALLEY HIGH SCHOOL

JC 2

PRELIMINARY EXAMINATION

CANDIDATE
NAME

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CENTRE
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CLASS

20J

INDEX
NUMBER

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H2 BIOLOGY

9744/02

Paper 2 Structured Questions

13 September 2021

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
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Total	

Answer **all** questions.

- 1 Guard cells are cells surrounding each stoma found on the underside of leaves in plants. Opening of stoma is regulated by an increase in turgidity of guard cells. Fig. 1.1 shows a series of processes that occur at the cell surface membrane of a guard cell.

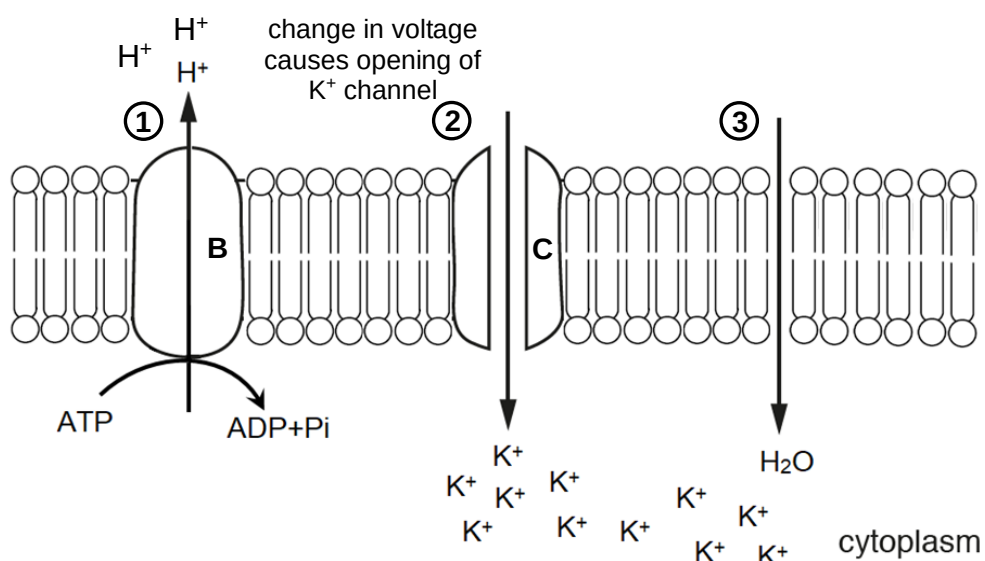


Fig. 1.1

- (a) Describe how the arrangement of phospholipids allows for the maintenance of different concentrations of H^+ and K^+ ions across the cell surface membrane. [2]

- (b) Voltage is the electrical potential difference across the membrane which is influenced by the concentration of ions.

With reference to Fig. 1.1, outline how the change in voltage results in the opening of a stoma.

[3]

- (c) Describe **two** structural similarities between structures of **B** and **C** that enable them to perform their functions.

[2]

A study was carried out to investigate the membrane permeability of an artificial membrane that lacked proteins.

Table 1.1 shows the results of this study and the radius of each substance tested.

Table 1.1

substance	membrane permeability / nm s ⁻¹	radius of one molecule or ion / nm
sodium ion	10 ⁻⁵	0.10
oxygen	>10 ⁵	0.12
glycerol	10 ¹	0.31
glucose	10 ⁻³	0.36
albumin protein	0	3.55

- (d) Describe how properties of substances affect membrane permeability and support your answer with evidence from Table 1.1. [3]

[Total: 10]

- 2 Starch is a storage polysaccharide found in plants. Fig. 2.1 shows a representation of a starch granule.

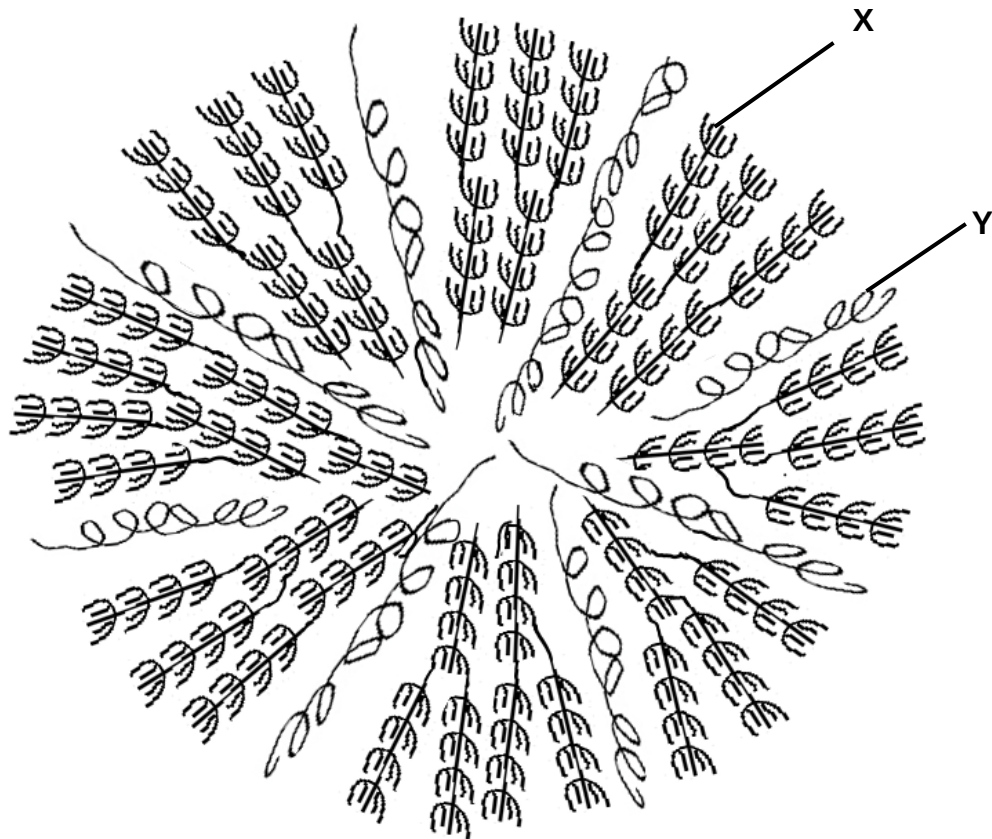


Fig. 2.1

- (a) Identify molecules X and Y.

[2]

- (b) Explain how the structural features of starch visible in Fig. 2.1 are related to its storage role in plants.

[2]

- (c) Explain how the structure of molecule **Y** differs from that of haemoglobin. [2]

Cellulose is another polysaccharide found in plants.

- (d) In the space below, show how a glucose molecule is released from part of a cellulose chain consisting of three monomers. [3]

[Total: 9]

- 3 (a) State the function of DNA and describe how its properties allow it to perform this function. [3]

Fig. 3.1 shows the process by which DNA is replicated.

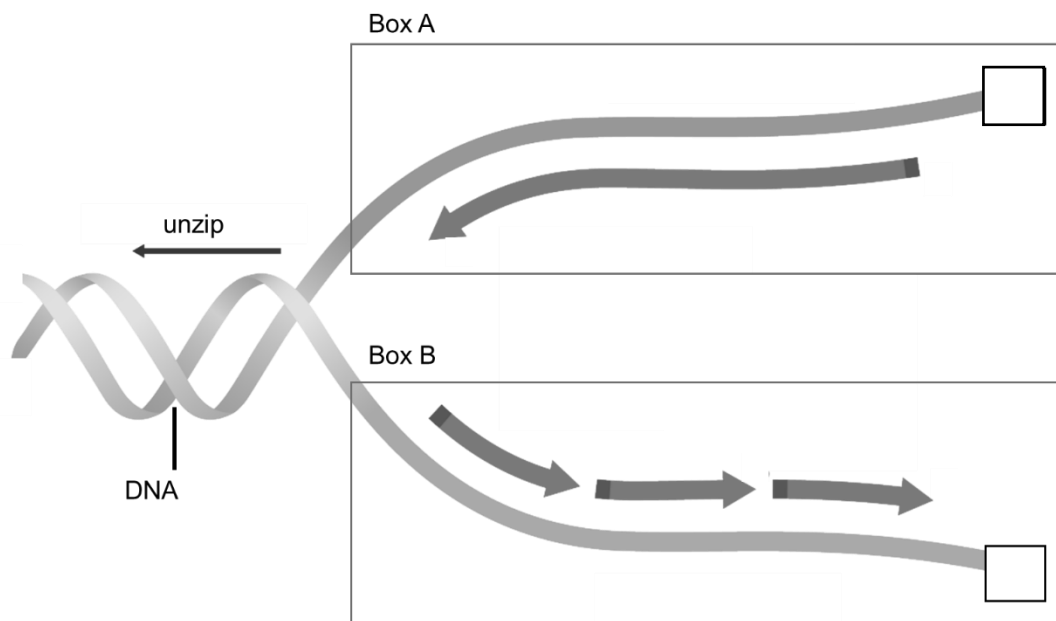


Fig. 3.1

- (b) (i) Label the 3' and 5' ends of DNA in the boxes provided in Fig. 3.1. [1]
 (ii) Account for the difference in how DNA is replicated in Box A and Box B. [3]

Antibiotics are commonly used in modern medicine to kill bacterial pathogens. Quinolones are a type of antibiotics and its mechanism of action is shown in Fig. 3.2.

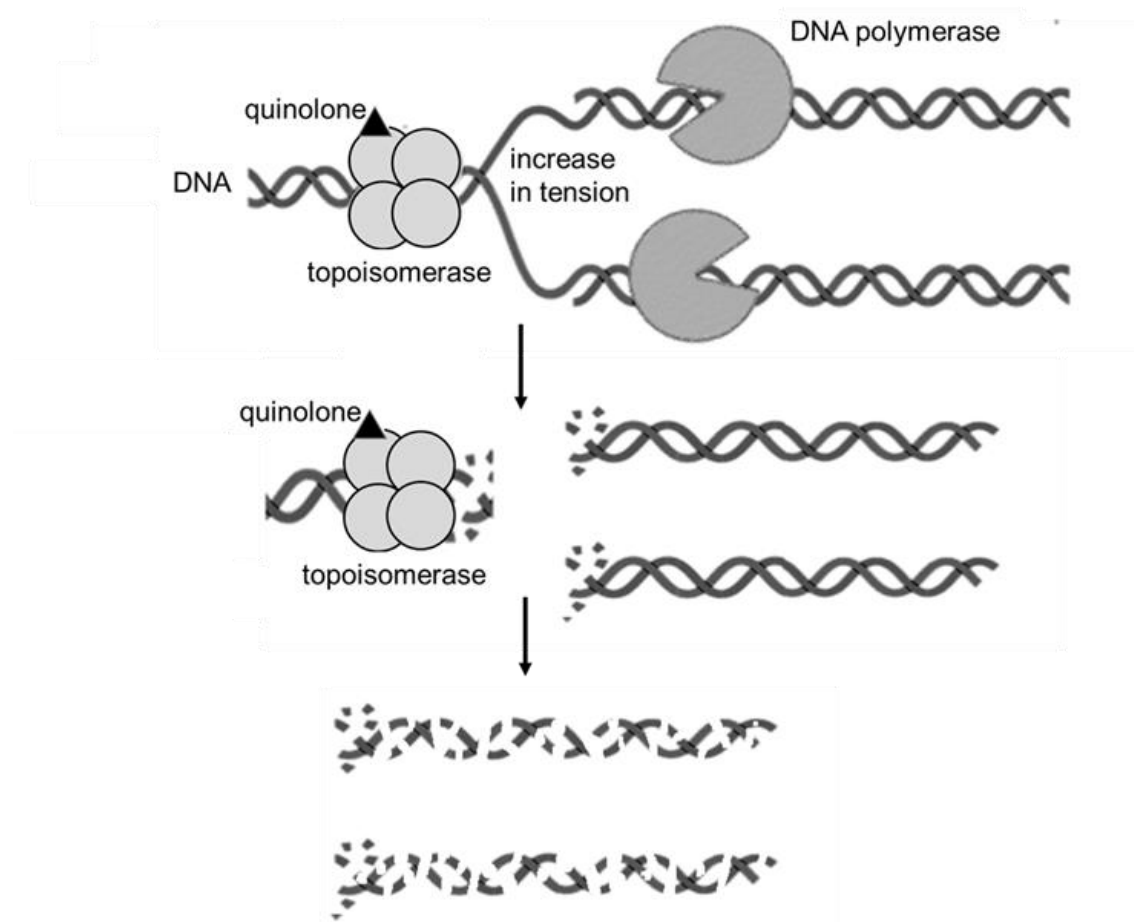


Fig. 3.2

- (c) With reference to Fig. 3.2 and your knowledge of DNA replication, describe how quinolone leads to bacterial cell death. [3]

[Total: 10]

Question 4 starts on page 10

- 4 *BRCA1* gene is a tumour suppressor gene that is involved in DNA repair and transcriptional regulation. Inherited mutations in the *BRCA1* gene are mainly linked to increased breast and ovarian cancer risks. It is estimated that about 3% of breast cancers (7500 women per year) and about 10% of ovarian cancers (2000 women per year) result from inherited mutations in the *BRCA1* gene.

Table 4.1 shows the analysis of mutation types in the *BRCA1* gene in a sample of 53 individuals with heritable **breast cancer**.

Table. 4.1

mutation type	number of samples with mutation	percentage
substitution nonsense	6	11.3
substitution missense	28	52.8
insertion in frame	0	0.0
insertion frameshift	5	9.4
deletion in frame	1	1.9
deletion frameshift	7	13.2
other	6	11.3
total unique samples	53	

Table 4.2 shows the analysis of mutation types in the *BRCA1* gene in a sample of 89 individuals with heritable **ovarian cancer**.

Table. 4.2

mutation type	number of samples with mutation	percentage
substitution nonsense	24	27.0
substitution missense	29	32.6
insertion in frame	0	0.0
insertion frameshift	7	7.9
deletion in frame	0	0.0
deletion frameshift	21	23.6
other	8	9.0
total unique samples	89	

Adapted from: Liang *et al*, International Journal of Molecular Medicine (2018)

- (a) (i) Calculate the total number of individuals per year who develop **breast** and **ovarian** cancers due to substitution mutations.

Show your working clearly.

[2]

- (ii) The deletion in frame of *BRCA1* gene in the sample for **breast** cancer resulted in a BRCA1 protein that is 23 amino acids longer.

Explain how this occurs.

[2]

Mutation screening, using gel electrophoresis, was performed to compare the nucleotide sequences of a normal and a mutated *BRCA1* gene.

Fig. 4.1 shows part of the sequence from the gel electrophoresis.

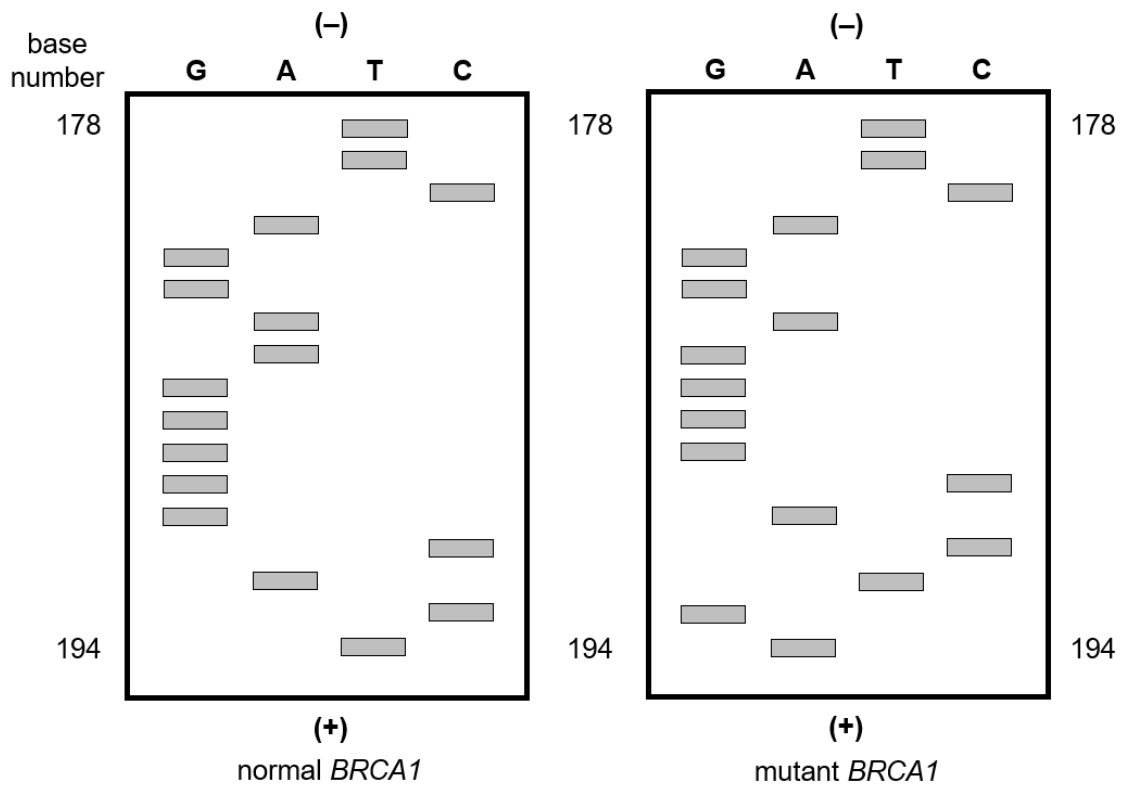


Fig. 4.1

- (b) Using the information provided in Fig. 4.1, describe precisely the mutation and explain how it results in the formation of cancer. [4]

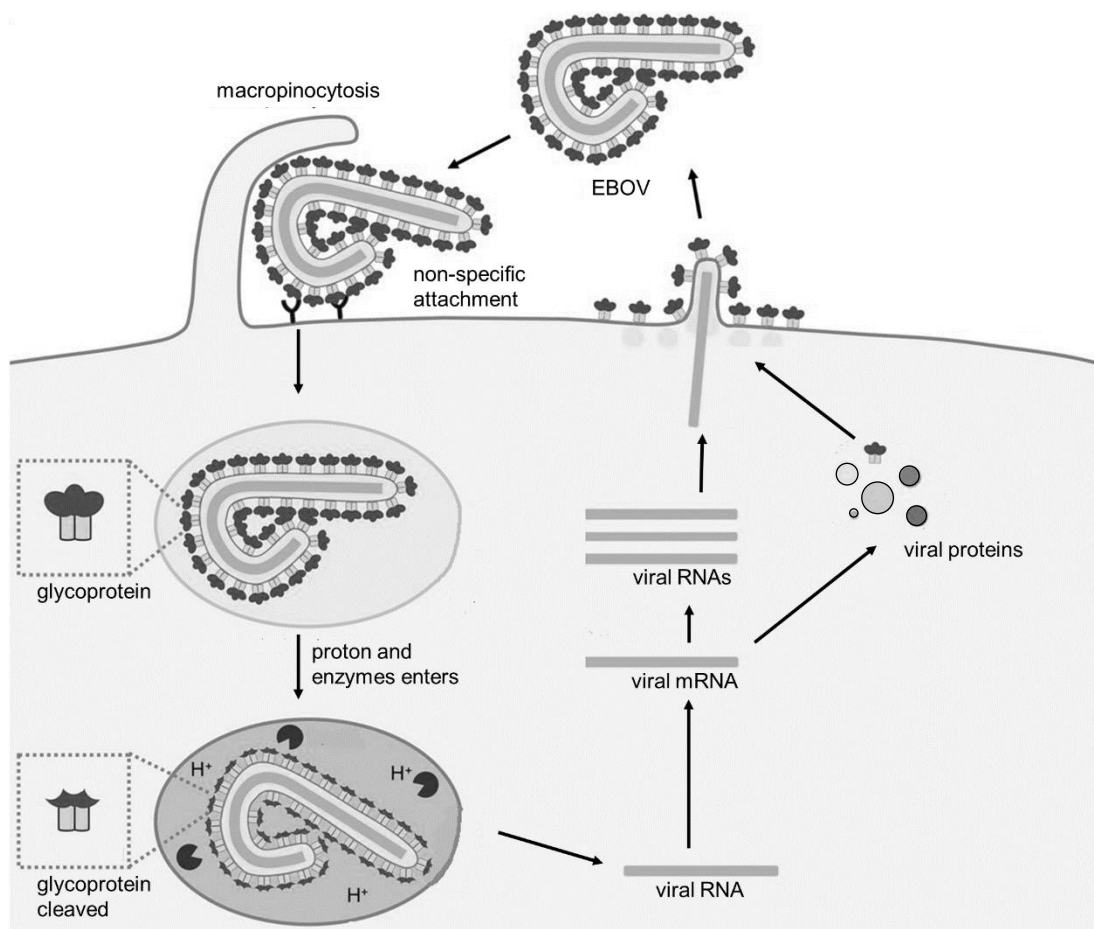
- (c) Other than tumor suppressor genes, oncogenes have a key role in cancer.

Explain what is meant by an oncogene in the context of cancer formation. [2]

[Total: 10]

- 5 Ebola is a rare yet deadly infectious disease. The virus that causes Ebola is known as the Ebola virus (EBOV). EBOV is an enveloped virus containing a cylindrical nucleocapsid that encloses a RNA genome and a viral enzyme.

Fig. 5.1 shows part of the reproductive cycle of EBOV.



Adapted from Sullivan *et al*, The Journal of Immunology, 201 (3) 833-842, August 1 (2018)

Fig. 5.1

- (a) Explain why viruses, such as EBOV, are obligate parasites. [1]

(b) With reference to Fig. 5.1 and your knowledge of HIV,

(i) distinguish between the entry of EBOV to that of HIV into its host cell, [3]

(ii) suggest the identity of the enzyme present in the nucleocapsid of EBOV and explain why this enzyme is needed to form new virions. [3]

(c) EBOV acquires its viral envelope using the same process as HIV.

Describe how EBOV acquires its viral envelope. [3]

[Total: 10]

- 6** The formation of the *lac* operon was an important adaptation of prokaryotes in response to its environment. This organisation of genome contributed to an increased evolutionary advantage for prokaryotes.

(a) Describe how the *lac* operon contributed to increased evolutionary advantage. [4]

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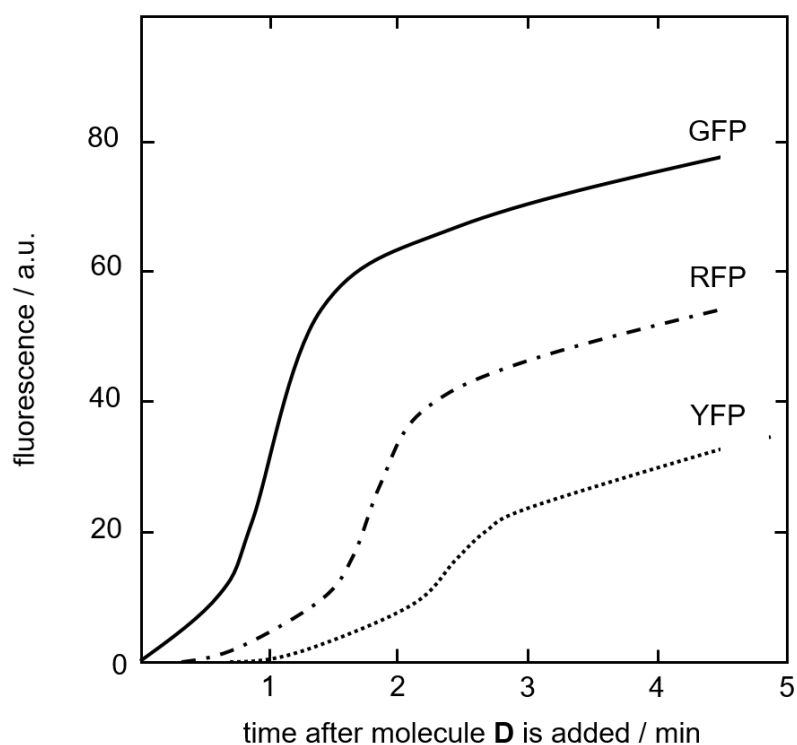
In an investigation on the effect of gene position on rate of expression, the structural genes in the *lac* operon were replaced by genes coding for green fluorescence protein (GFP), red fluorescence protein (RFP) and yellow fluorescence protein (YFP). This allowed for the observation of expression of structural genes in the *lac* operon.

Fig. 6.1 shows the structure of the modified *lac* operon.



Fig. 6.1

The modified *lac* operon was inserted into a prokaryotic cell. This prokaryotic cell has no genome and contains only molecular machinery for gene expression and an excess of catabolite activator protein and *lac* repressor protein. The operon was induced to turn on by adding molecule **D**. Fig. 6.2 shows the results of the experiment.



Adapted from: Hussein *et al*, PNAS (2011)

Fig. 6.2

- (b) Describe how molecule **D** induces the synthesis of GFP protein. [3]

- (c) (i) With reference to Fig. 6.2, explain the differences in fluorescence of GFP and YFP due to induction of modified *lac* operon by molecule **D**. [2]

- (ii) Suggest the significance of this finding on GFP in (c)(i) to the uptake of lactose by prokaryotes. [1]

[Total: 10]

- 7 In domestic pigeons, the presence of feathers on their head is known as a crest. This is determined by a gene found on an autosomal chromosome.

In domestic pigeons, the sex chromosomes are Z and W, and the heterogametic sex is the female. Thus, the female has one Z and one W chromosome (ZW) while the male has two Z chromosomes (ZZ). The gene controlling plumage colour of the pigeon is found on chromosome Z.

Table 7.1 shows the phenotypes of 13 offspring from the cross of an ash-red female pigeon with no crest and an ash-red male pigeon with crest.

Table 7.1

offspring number	sex	colour	presence of crest
1	male	ash-red	no crest
2	male	ash-red	no crest
3	male	ash-red	no crest
4	male	ash-red	no crest
5	male	ash-red	no crest
6	male	ash-red	no crest
7	male	ash-red	no crest
8	female	ash-red	no crest
9	female	ash-red	no crest
10	female	blue	no crest
11	female	blue	no crest
12	female	blue	no crest
13	female	blue	no crest

- (a) State the dominant allele for both the gene loci controlling presence of crest and plumage colour. [2]

- (b)** Using the following symbols to represent the alleles, draw a genetic diagram to explain the results of the cross.

key

A – ash-red plumage

B – blue plumage

D – no crest

d – crest

[4]

χ^2 value for the cross can be calculated using the following below.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Where Σ = sum of

O = observed value

E = expected value

- (c) (i)** Using the genetic diagram in **(b)** and Table 7.1, calculate the χ^2 value for the cross.

[2]

Critical values of this χ^2 distribution are shown in Table 7.2.

Table 7.2

degrees of freedom	probability, p						
	0.95	0.80	0.50	0.20	0.10	0.05	0.01
1	0.004	0.06	0.46	1.64	2.71	3.84	6.64
2	0.10	0.45	1.39	3.22	4.61	5.99	9.21
3	0.35	1.01	2.37	4.64	6.25	7.82	11.35
4	0.71	1.65	3.36	5.59	7.78	9.49	13.28

- (ii) Using the calculated χ^2 value in (c)(i) and Table 7.2, explain what conclusion can be drawn from the results.

[2]

[Total: 10]

Question 8 starts on page 24

- 8 Fig 8.1 shows part of the process of aerobic respiration in *S. aureus*, a gram positive bacteria.

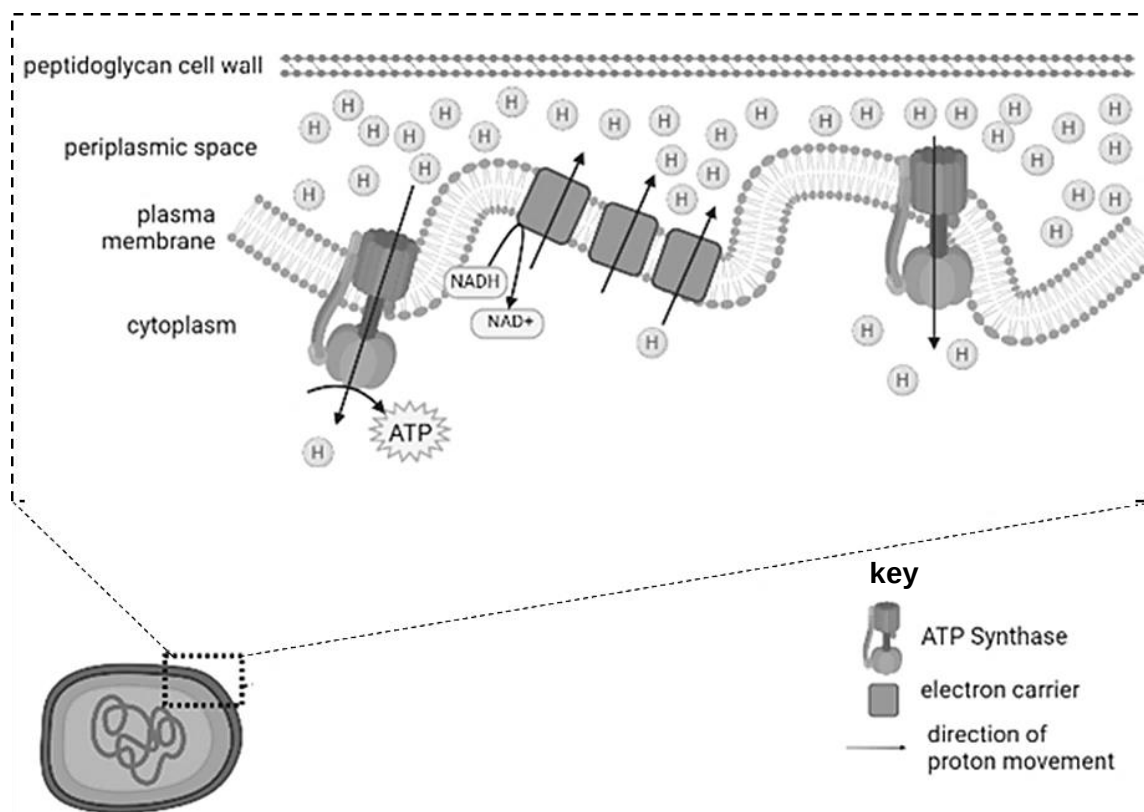


Fig 8.1

- (a) Glucose enters bacteria cells via glucose transporters found on the plasma membrane. Glucose then undergoes glycolysis.

Outline briefly the process of glycolysis for one molecule of glucose.

[3]

- (b)** Distinguish between the process shown in Fig 8.1 and non-cyclic photophosphorylation in plant cells. [4]

- (c)** Describe the role of NAD in aerobic respiration. [2]

[Total: 9]

- 9 (a) Describe how micro-evolution is related to macro-evolution. [2]

Modern varieties of wheat have developed from numerous hybridisation events between different species of wild grasses. Fig. 9.1 shows some of the possible steps that are believed to have been involved in the development of bread wheat, *Triticum aestivum*.

The letters **A**, **B** and **C** represent three different sets of seven chromosomes. For example, in *Triticum urartu* has two **A** and therefore possesses fourteen chromosomes.

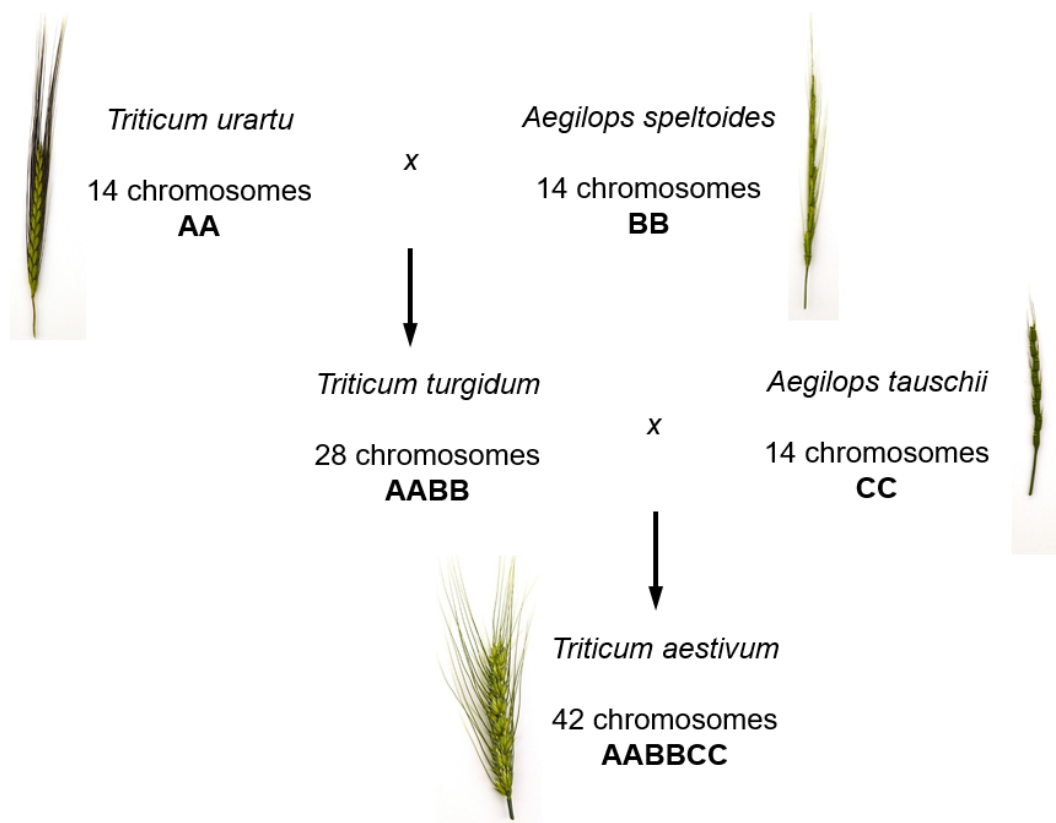


Fig. 9.1

- (b) With reference to Fig. 9.1, explain how the new species of bread wheat, *Triticum aestivum*, formed from *Triticum turgidum*. [4]

- (c) **A**, **B** and **C** contain common genes that code for the same proteins essential to the survival of modern wheat. Recessive alleles of these genes were found to be more prevalent in the population of *Triticum aestivum* compared to the other varieties.

Explain why recessive alleles present in **A** may be better preserved in the population of *Triticum aestivum* than in that of *Triticum urartu*, [2]

- (d) Explain why genetic variation is important to the survival of a species. [2]

[Total: 10]

- 10** Contact tracing is a key process to slow down the spread of COVID-19.

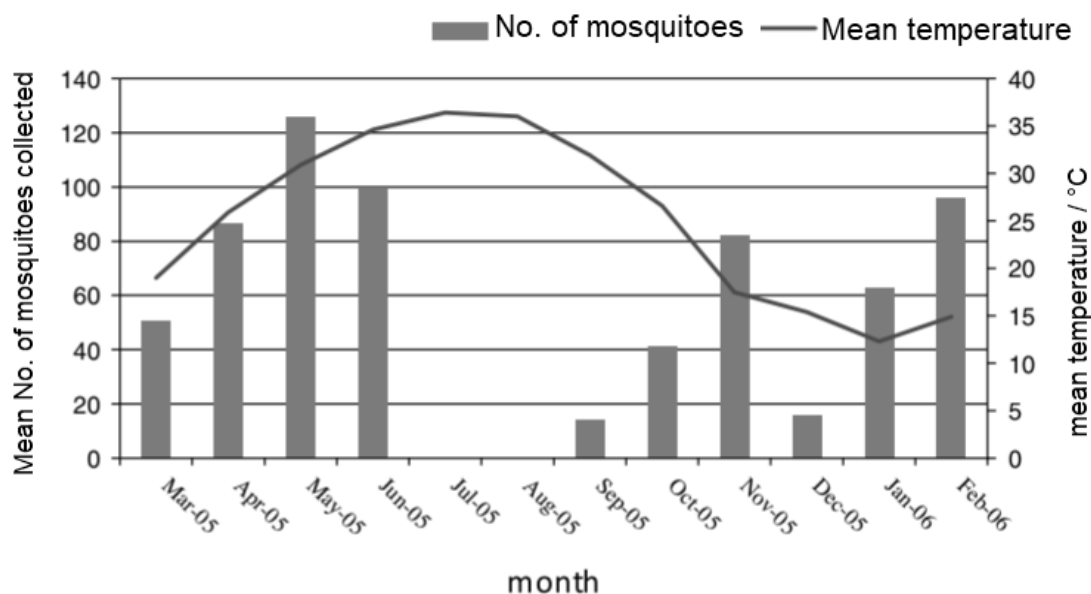
Antigen test and antibody test detect the presence of COVID-19 virus through the presence of the COVID-19 antigen or COVID-19 antibody respectively in blood samples of individuals who are suspected to be infected with the virus. These tests are done in the effort to map the transmission network of the virus amongst individuals so that individuals who are capable of transmitting the virus can be quickly quarantined.

- (a)** Describe how an antigen presenting cell presenting the COVID-19 antigen led to the production of COVID-19 antibodies. [4]

- (b)** Suggest why both antigen test and antibody test are necessary in the process of contact tracing to map the disease transmission network. [3]

[Total: 7]

- 11** Insects are the most dominant group of organisms on the planet in terms of species richness, abundance and biomass. Global warming has a marked influence on the physiology of insects. One such insect is the mosquito. Fig. 11.1 shows the effect of temperature on seasonal abundance of mosquitoes in the tropics.



Source: Mosquito fauna (Diptera: Culicidae) of the Eastern Region of Saudi Arabia and their seasonal abundance, *Journal of King Saud University - Science* · January 2010

Fig. 11.1

(a) With reference to Fig. 11.1,

- (i)** explain the effect of temperature on the mean number of mosquitoes between March and May 2005, [2]

- (ii)** identify the maximum temperature at which the mosquitoes can survive. [1]

- (iii)** explain why global warming can lead to spread of dengue beyond the tropics. [2]

[Total: 5]

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